

Lab Exercise #7: Remove Odd Numbers From An Array

1 Overview

In this lab, write a C program which takes an array of integers, counts then number of even elements and returns the number of evens and prints the array with only the even entries.

Your program should

- Request the user to input the size of an 1 dimensional array and read in the size.
- Request the user to put in the array and read in the array.
- Create a function `RemoveOdds()` that has three parameters: the array size and two integer arrays.
- Function `RemoveOdds()` should fill the second array with the even elements of the first array and return the number of elements in the second array.
- The program should print out the number of evens in the original matrix and the new matrix with only the even elements.

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #7 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #7 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below.

```
Lab #8: Mark Schwitzerlett and Jacobi

Enter the size of your array: 9

Input your array: 21 34 52 68 77 89 90 101 114
The number of evens is 5

34, 52, 68, 90, 114

...Program finished with exit code 0
Press ENTER to exit console.[]
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named `Lab#7_FirstName_LastName.c`

Due date: The lab is due at the end of your lab period, either Monday, March 30th or Tuesday, March 31st.

Be sure to save a copy of your .c file.